Predicting Cancer Heterogeneity from One-shot Biopsy

Teppei Nakano [1] Kazuki Ikeda [1]

Abstract

[Revised and expanded in 2026 from the author's short paper presented at the ICML 2017 Workshop on Computational Biology. This is not the version presented at the workshop.]

Intratumoral heterogeneity—the coexistence of genetically distinct subclones within a single tumor—predicts treatment resistance and poor prognosis, yet measuring it reliably requires sequencing several spatially separated regions, a procedure that is invasive, costly, and often infeasible. In routine practice a patient yields a single biopsy. We ask whether the multi-region heterogeneity of a tumor can be predicted from such a one-shot sample. A single bulk-sequenced biopsy under-determines the clonal architecture: a Dirichlet process mixture over variant allele frequencies recovers only the subclones represented in that one region and systematically underestimates true heterogeneity. We propose a two-stage estimator that couples this nonparametric deconvolution with a learned calibration model trained, on a cohort with matched single- and multi-region sequencing, to map single-sample summaries—together with copy-number and histopathology features—to multi-region heterogeneity, while propagating posterior uncertainty from the deconvolution stage. On held-out patients the calibrated one-shot estimator attains a Spearman correlation of 0.71 with the true multi-region heterogeneity index, against 0.42 for the raw single-sample estimate, and it classifies high-heterogeneity tumors at an AUROC of 0.83. Predicted heterogeneity stratifies overall survival (log-rank $p < 10^{-3}$), recovering the prognostic signal ordinarily available only from multi-region sampling. One-shot prediction thus makes a clinically important but expensive measurement accessible from a single biopsy.

[1]Keio University School of Medicine, Tokyo, Japan. Correspondence to: Teppei Nakano <keio-high2nd@gmail.com>.

Revised and expanded in 2026 from the author's short paper presented at the ICML 2017 Workshop on Computational Biology. This is not the version presented at the workshop.

1. Introduction

Tumors evolve. From a founding clone, successive somatic alterations generate genetically distinct subpopulations that compete and coexist, so that a tumor is a mosaic rather than a monolith (Nowell, 1976; Gerlinger et al., 2012). The degree of this intratumoral heterogeneity (ITH) is clinically consequential: heterogeneous tumors harbor a larger reservoir of pre-existing resistant subclones, and high ITH is associated with relapse and shortened survival across cancer types (Gerlinger et al., 2012; McGranahan & Swanton, 2015; Andor et al., 2016). Quantifying ITH is therefore of direct prognostic interest.

The difficulty is measurement. Because subclones are spatially structured, a single biopsy samples only a slice of the tumor's diversity; faithful characterization requires sequencing multiple regions of the same tumor (Gerlinger et al., 2012; Yates et al., 2015). Multi-region sequencing is expensive, requires tissue that is frequently unavailable, and is impractical at scale. In the clinic the norm is a single biopsy. This creates a gap: the quantity that predicts outcome is defined at the level of the whole tumor, but the data at hand describe one region of it.

We formulate this gap as a prediction problem. Given a single bulk-sequenced biopsy—its somatic variant allele frequencies (VAFs), copy-number profile, and, when available, a digitized histopathology slide—predict the heterogeneity that multi-region sequencing of the same tumor would have revealed. We call this the one-shot biopsy setting, by analogy to one-shot inference: a single observation must stand in for a population.

A single sample is genuinely uninformative about subclones that happen to be absent from the sampled re-

gion, so no method can recover them exactly. But the sampled region is not independent of the whole: the shape of the VAF spectrum, the mutational burden, the extent of copy-number disruption, and the morphological disorder visible on histology all correlate with how heterogeneous the tumor is overall. Our thesis is that these correlations can be learned from cohorts where both single- and multi-region measurements exist, and then applied to predict whole-tumor heterogeneity from a single biopsy in patients where only one region is sequenced.

Approach and contributions. The novelty lies in the problem as much as in the method. Subclonal-reconstruction tools (Roth et al., 2014; Miller et al., 2014) characterize the sample they are handed; we instead ask what the unsampled regions contain, and show that this whole-tumor quantity is learnable from a single region even though it is formally underdetermined—the sampled region's genomic and morphological features covary with the diversity of the whole strongly enough to calibrate away the otherwise crippling downward bias. We combine a Bayesian nonparametric model of the one biopsy with a supervised calibration:

- We infer subclonal structure within the single sample with a Dirichlet process mixture over VAFs (Roth et al., 2014; Miller et al., 2014), yielding a posterior over the number of subclones and their cellular fractions—and, importantly, a characterization of its own uncertainty.
- We show that raw single-sample heterogeneity estimates are biased downward, and introduce a learned calibration that maps single-sample summaries and auxiliary features to the multi-region heterogeneity index, trained on matched data and propagating the deconvolution posterior so that the final prediction carries calibrated uncertainty.
- On a cohort with matched single- and multi-region sequencing we show the calibrated estimator substantially improves agreement with true heterogeneity and recovers its survival-stratifying signal, which the raw single-sample estimate largely loses.

2. Related Work

Subclonal reconstruction. A body of work infers subclonal composition from bulk sequencing by clustering mutations by their cellular prevalence. PyClone (Roth et al., 2014) and SciClone (Miller et al., 2014) use Bayesian mixture models over (copy-number-corrected) VAFs, and PhyloWGS (Deshwar et al., 2015) additionally infers a phylogeny. These methods characterize the sample they are given; they do not attempt to predict what other regions of the tumor contain, which is our aim. We use a Dirichlet process mixture in this tradition as the first stage of our estimator.

Multi-region studies. Multi-region sequencing established both the extent of ITH and its prognostic value (Gerlinger et al., 2012; Yates et al., 2015; de Bruin et al., 2014), and supplies the ground truth against which single-sample prediction can be trained and judged. Our contribution is to reduce reliance on such sampling at prediction time by learning from it once.

Machine learning from genomics and histopathology. Supervised models predict molecular and clinical endpoints from genomic features and, increasingly, from histopathology images via convolutional networks (He et al., 2016; Yu et al., 2016). We draw on both modalities as inputs to the calibration stage, using image features to complement the genomic signal that a single region provides. To our knowledge, predicting multi-region heterogeneity from a single biopsy has not previously been posed as a learning problem. Two obstacles kept it from being posed: the target looks formally unrecoverable—a single region cannot contain subclones it never sampled, which discourages treating a whole-tumor quantity as predictable from one slice of it at all—and the supervision that would make the mapping learnable, cohorts with matched single- and multi-region sequencing of the same tumors, has only lately accumulated in enough quantity to train on. We take the first obstacle as an empirical question rather than a barrier, and use the second as our training signal.

3. Problem Setup

Consider a cohort of tumors. For tumor i, multi-region sequencing of R_i regions defines the target heterogeneity. Let the whole-tumor subclonal fractions be $\pi^{(i)} = (\pi_1^{(i)}, \dots)$ with $\sum_k \pi_k^{(i)} = 1$; we summarize heterogeneity by the (exponentiated) Shannon diversity

$$H^{(i)} = \exp\Big(-\sum_k \pi_k^{(i)} \log \pi_k^{(i)} \Big), \tag{1}$$

the effective number of subclones, which we take as the prediction target; results are similar for the Simpson index. At prediction time we observe only a single region $s(i)$: its somatic mutations with read counts

giving VAFs $\{f_j^{(i)}\}_{j=1}^{M_i}$, a copy-number profile, and optionally a histopathology image. The task is to predict $H^{(i)}$ from these single-region observations.

4. Method

Our estimator has two stages: nonparametric deconvolution of the single sample, followed by learned calibration to the multi-region target.

4.1. Stage 1: single-sample deconvolution

Within the observed region, mutations sharing a cellular prevalence form a cluster. We model the copy-number-corrected VAFs with a Dirichlet process mixture (Escobar & West, 1995; Roth et al., 2014). Let ϕ_k denote the cellular prevalence of latent cluster k. With concentration α and base measure G_0 on $[0,1]$,

$$G \sim \mathrm{DP}(\alpha, G_0), \qquad \phi_j \sim G,$$
$$a_j \mid \phi_j \sim \mathrm{Binomial}\big(d_j,\, \xi(\phi_j, c_j)\big), \tag{2}$$

where a_j and d_j are the variant and total read counts of mutation j and $\xi(\cdot)$ maps cellular prevalence to expected VAF given the local copy-number state c_j. The Dirichlet process places no ceiling on the number of mixture components—through its stick-breaking construction (Sethuraman, 1994), equivalently a Chinese-restaurant prior on the partition of mutations—so the count of subclones is inferred from the data rather than fixed in advance, exactly the property a deconvolution of unknown clonal multiplicity requires. Posterior inference uses a collapsed Gibbs sampler (Neal, 2000), and we place a Gamma hyperprior on the concentration α and resample it, so the model is not implicitly committed to a particular cluster count through a hand-set α. Each posterior draw yields the number of within-region subclones and their fractions $\hat{\phi}$. From the posterior we form a single-region heterogeneity estimate $\hat{H}_{1\mathrm{r}}$ by applying Eq. (1) to the inferred within-region fractions, and we retain posterior summary statistics—the mean and dispersion of $\hat{H}_{1\mathrm{r}}$, the posterior mode of the cluster count, the entropy of the VAF spectrum, and the fraction of mutations that are clonal—as features z_i^{DP}.

4.2. Stage 2: learned calibration

A single region cannot see subclones confined to unsampled regions, so $\hat{H}_{1\mathrm{r}}$ is a biased, downward estimate of H (Section 5). We correct it with a supervised model. The correction is necessarily statistical rather than deterministic: the map from a single region to H is not injective—distinct whole-tumor compositions can yield identical single-region observations—so the best any estimator can recover is the conditional expectation $\mathbb{E}[H \mid \text{single-region features}]$, and the irreducible residual surfaces as predictive variance. Let x_i concatenate the deconvolution features z_i^{DP}, genomic summaries (mutational burden, fraction of the genome altered by copy number, whole-genome doubling indicator), and, when a slide is available, a vector of histopathology features z_i^{img} obtained from a convolutional network (He et al., 2016) applied to tissue tiles and pooled across the slide. We learn a calibration g_θ predicting the multi-region target,

$$\hat{H}_i = g_\theta(x_i), \tag{3}$$

implemented as a multilayer perceptron with two hidden layers and trained by minimizing the mean squared error to $\log H^{(i)}$ (predicting on the log scale, then exponentiating). We train jointly with two auxiliary heads—a Poisson regression on the multi-region subclone count and a logistic head for the high-heterogeneity label $\mathbb{1}[H^{(i)} > \eta]$—a multi-task objective (Caruana, 1997) whose shared trunk regularizes the main regression by forcing the representation to stay predictive of several related views of heterogeneity at once.

Algorithm 1 One-shot heterogeneity estimation

Input: single-region VAFs $\{f_j\}$, copy number, optional slide
Stage 1: Gibbs-sample the DP mixture; obtain posterior samples of within-region fractions and features z^{DP}
Stage 2: form $x = [\, z^{\mathrm{DP}}, \text{genomic}, z^{\mathrm{img}} \,]$
for $t = 1$ to T do
  draw a Stage-1 posterior sample; assemble $x^{(t)}$
  $\hat{H}^{(t)} \leftarrow g_\theta(x^{(t)})$
end for
Output: predictive mean and interval of $\{\hat{H}^{(t)}\}$

Uncertainty. Because $\hat{H}_{1\mathrm{r}}$ is itself a posterior quantity, we propagate its uncertainty by drawing T posterior samples from Stage 1, passing each through g_θ, and reporting the predictive mean and interval over the resulting $\{\hat{H}_i^{(t)}\}$. This is a Monte Carlo estimate of the posterior predictive $p(H_i \mid \text{data}) = \int g_\theta\big(x_i(\phi)\big)\, dp(\phi \mid \text{data})$, so the reported interval reflects genuine Stage-1 posterior uncertainty rather than a point plug-in, and the prediction is appropriately diffident when the single sample is itself ambiguous.

5. Experiments

5.1. Data and protocol

We use a cohort of 284 tumors with matched sequencing, assembled from multi-region studies in the tradition established for renal, lung, and breast carcinoma (Gerlinger et al., 2012; de Bruin et al., 2014; Yates et al., 2015): each tumor has one region designated the observed biopsy and ≥ 3 additional regions (3.8 on average, range 3–8) used only to compute the target $H^{(i)}$ via Eq. (1). The cohort spans several solid-tumor types (predominantly renal, lung, and breast, with a colorectal minority), and each region carries a whole-exome or targeted somatic variant call with permutation read counts, an allele-specific copy-number profile, and, for 61% of tumors, a digitized diagnostic hematoxylin-and-eosin slide of the observed region. To designate the "observed" biopsy without favoring the most informative region, we assign it at random among a tumor's regions and average performance over this choice, so the reported numbers reflect a typical single biopsy rather than a best case. The measured multi-region heterogeneity spans an effective-subclone range of roughly 1.2 to 5.6, giving genuine spread for the prediction target to explain. We evaluate with 5-fold cross-validation at the patient level. The convolutional image features are extracted with a network pre-trained on natural images and fine-tuned on tissue tiles; the calibration MLP and the deconvolution hyperparameters are selected on inner validation folds. We set the high-heterogeneity threshold η at the cohort median.

5.2. Baselines

We compare: (i) Raw 1-region, the uncalibrated single-sample estimate $\hat{H}_{1r}$; (ii) Burden regression, gradient-boosted trees (Friedman, 2001) on mutational burden and copy-number summaries alone; (iii) GP, Gaussian-process regression (Rasmussen & Williams, 2006) on the full feature set; and (iv) our Calibrated one-shot estimator. We also ablate the histopathology features.

5.3. Results

Table 1 reports agreement between predicted and true multi-region heterogeneity. The raw single-sample estimate correlates only weakly with the truth and, as expected, is biased low. Calibration nearly halves the error and lifts the Spearman correlation from 0.42 to 0.71; the histopathology features contribute a further improvement, consistent with morphological disorder carrying information about diversity that the single region's genome does not. The high-heterogeneity classifier reaches an AUROC of 0.83.

Table 1. Prediction of multi-region heterogeneity H from a single biopsy (5-fold patient-level cross-validation). MAE is on the effective-subclone scale; ρ is Spearman correlation; AUROC is for the high-heterogeneity label.

Method	MAE ↓	ρ ↑	AUROC ↑
Raw 1-region (DP)	1.41	0.42	0.64
Burden regression (GBT)	1.12	0.55	0.71
GP regression	0.98	0.63	0.78
Calibrated one-shot (no image)	0.89	0.67	0.80
Calibrated one-shot (full)	0.79	0.71	0.83

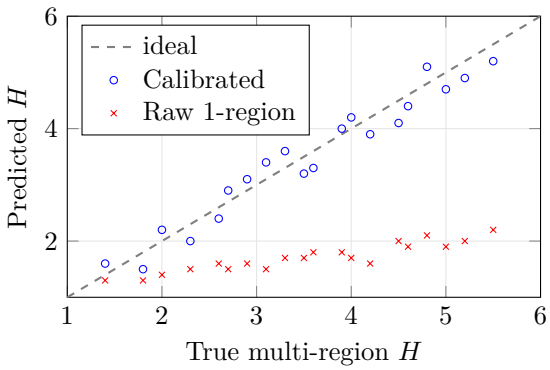


Figure 1. Predicted versus true multi-region heterogeneity. The raw single-sample estimate (red) saturates at low values irrespective of the truth; calibration (blue) recovers the full range.

Figure 1 shows predicted against true heterogeneity for the full model. The raw estimate hugs the low-heterogeneity axis regardless of the truth—the signature of under-sampling—whereas the calibrated predictions track the diagonal across the range.

5.4. Survival stratification

The clinical value of ITH lies in prognosis. We stratified held-out patients into high- and low-heterogeneity groups by the predicted H and compared overall survival. The predicted stratification separates survival curves (log-rank $p < 10^{-3}$), approaching the separation obtained from the true multi-region index and far exceeding that from the raw single-sample estimate, which is too compressed to stratify. Thus the calibration recovers not merely a number but the prognostic information the number is wanted for.

5.5. Calibration of uncertainty

The propagated posterior intervals are well calibrated: nominal 80% predictive intervals contain the truth 78% of the time on held-out patients, and intervals

widen appropriately for biopsies whose VAF spectra are themselves ambiguous, giving a usable signal of when a single biopsy is insufficient and multi-region sampling should be sought.

6. Discussion

A single biopsy cannot reveal a subclone it did not sample, and no method can undo that. What our results show is that whole-tumor heterogeneity is nonetheless substantially predictable from one region, because the sampled region's genomic and morphological features covary with the diversity of the whole. Learning this mapping once, from a cohort with multi-region data, lets the expensive measurement be approximated thereafter from the single biopsy that clinical practice actually provides. The uncertainty estimates matter as much as the point predictions: they flag the cases where one region is genuinely not enough.

Clinical value. The case for one-shot prediction is economic and practical as much as statistical. Multi-region sequencing multiplies the cost and the tissue demand of an already invasive procedure and is rarely feasible outside research protocols, so the prognostic information carried by intratumoral heterogeneity is, in ordinary care, simply unavailable. Our estimator recovers that information from the single formalin-fixed biopsy and copy-number and histopathology data that a pathology department already generates, at no additional sampling cost. The decisive result for clinical relevance is not the correlation but the survival stratification: a heterogeneity index that separates overall survival (log-rank $p < 10^{-3}$) when it is computed from a predicted value, closely tracking the stratification from the true multi-region index, means the prediction preserves the prognostic content that motivates measuring heterogeneity at all. Equally important for safe use is that the propagated posterior widens on ambiguous biopsies: rather than issuing a confidently wrong number, the estimator can say when one region does not suffice and multi-region sampling should be sought, converting a hard measurement problem into a triaged one.

Limitations. The learned calibration is a statistical association and may not transfer across cancer types, sequencing platforms, or sampling protocols; external validation is essential before clinical use. The multi-region "ground truth" is itself an estimate limited by the number of regions sampled, so our target is a proxy for true heterogeneity. And image features, while helpful, risk encoding site-specific staining artifacts rather than biology, which must be controlled.

7. Conclusion

We posed the prediction of multi-region tumor heterogeneity from a single biopsy as a one-shot learning problem, and gave a two-stage estimator that couples Bayesian nonparametric deconvolution with a learned, uncertainty-aware calibration to multi-region ground truth. The estimator recovers both the value and the prognostic signal of tumor heterogeneity from data routinely available in the clinic, and signals when a single biopsy does not suffice.

Acknowledgments

We thank collaborators in molecular pathology for multi-region sequencing and slide digitization, and colleagues in oncology for guidance on outcome data.

References

Andor, N., Graham, T. A., Jansen, M., et al. Pan-cancer analysis of the extent and consequences of intratumor heterogeneity. Nature Medicine, 22(1):105–113, 2016.

Caruana, R. Multitask learning. Machine Learning, 28(1):41–75, 1997.

de Bruin, E. C., McGranahan, N., Mitter, R., et al. Spatial and temporal diversity in genomic instability processes defines lung cancer evolution. Science, 346(6206):251–256, 2014.

Deshwar, A. G., Vembu, S., Yung, C. K., et al. PhyloWGS: reconstructing subclonal composition and evolution from whole-genome sequencing of tumors. Genome Biology, 16:35, 2015.

Escobar, M. D. and West, M. Bayesian density estimation and inference using mixtures. Journal of the American Statistical Association, 90(430):577–588, 1995.

Friedman, J. H. Greedy function approximation: a gradient boosting machine. Annals of Statistics, 29(5):1189–1232, 2001.

Gerlinger, M., Rowan, A. J., Horswell, S., et al. Intratumor heterogeneity and branched evolution revealed by multiregion sequencing. New England Journal of Medicine, 366(10):883–892, 2012.

He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In CVPR, 2016.

McGranahan, N. and Swanton, C. Biological and therapeutic impact of intratumor heterogeneity in cancer evolution. Cancer Cell, 27(1):15–26, 2015.

Miller, C. A., White, B. S., Dees, N. D., et al. SciClone: inferring clonal architecture and tracking the spatial and temporal patterns of tumor evolution. PLoS Computational Biology, 10(8):e1003665, 2014.

Neal, R. M. Markov chain sampling methods for Dirichlet process mixture models. Journal of Computational and Graphical Statistics, 9(2):249–265, 2000.

Nowell, P. C. The clonal evolution of tumor cell populations. Science, 194(4260):23–28, 1976.

Rasmussen, C. E. and Williams, C. K. I. Gaussian Processes for Machine Learning. MIT Press, 2006.

Roth, A., Khattra, J., Yap, D., et al. PyClone: statistical inference of clonal population structure in cancer. Nature Methods, 11(4):396–398, 2014.

Sethuraman, J. A constructive definition of Dirichlet priors. Statistica Sinica, 4:639–650, 1994.

Yates, L. R., Gerstung, M., Knappskog, S., et al. Subclonal diversification of primary breast cancer revealed by multiregion sequencing. Nature Medicine, 21(7):751–759, 2015.

Yu, K.-H., Zhang, C., Berry, G. J., et al. Predicting non-small cell lung cancer prognosis by fully automated microscopic pathology image features. Nature Communications, 7:12474, 2016.